On the Erdős-Turán Conjecture on Additive Bases

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Abstract

We explore the Erdős-Turán conjecture, postulating that for any additive basis of order 2 of the natural numbers, the representation function is unbounded. This document formalizes the axiomatic definitions and lemmas necessary for its resolution.

1 Introduction and Axiomatic Definitions

Let $\mathcal{A} \subseteq \mathbb{N}$ be a set of natural numbers.

The representation function of order 2, denoted $r_{\mathcal{A}}(n)$, counts the number of ways to express a natural number n as the sum of two elements in $\mathcal{A}$:

$$r_{\mathcal{A}}(n) = |\{(a,b) \in \mathcal{A} \times \mathcal{A} \mid a \leq b \text{ et } a+b=n\}|$$

Definition 1. *The set $\mathcal{A}$ is called an* ***additive basis of order 2*** *if there exists a constant $N_0 \geq 0$ such that for all $n \geq N_0$, $r_{\mathcal{A}}(n) \geq 1$.*

Theorem 1 (Conjecture d'Erdős-Turán)**.** *If $\mathcal{A}$ is an additive basis of order 2, then* $\limsup_{n\to\infty} r_{\mathcal{A}}(n) = \infty$.

2 Contextual Literature Research

A query to Arxiv.org revealed the following recent articles related to the topic:

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Title: Generalized additive bases, Konig's lemma, and the Erdos-Turan conjecture
Authors: Melvyn B. Nathanson
Summary:   Let A be a set of nonnegative integers. For every nonnegative integer n and positi
URL: https://arxiv.org/pdf/math/0302155v3
```

An interesting analogy can be drawn with the recently resolved Cap Set problem. In both cases, the study involves the absence or abundance of certain additive configurations (even sums or arithmetic progressions) in large sets. The polynomial methods, which were crucial for the Cap Set problem, might inspire new techniques to approach the Erdős-Turán representation function, particularly by employing the fundamental theorem of algebra over finite fields and additive Fourier analysis.

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3 Proof Strategy and Lemmas

We decompose the problem by studying the properties of sets whose representation function is bounded, aiming to reach a contradiction.

Lemma 1. *Let $\mathcal{A} \subseteq \mathbb{N}$. If $r_{\mathcal{A}}(n) \leq K$ for all n, then the generating series $f(z) = \sum_{a \in \mathcal{A}} z^a$, analytic on the unit disk $|z| < 1$, satisfies a restrictive boundary condition.*

Proof. Let $\mathcal{A}$ such that $r_{\mathcal{A}}(n) \leq K$. Consider the formal power series $f(z) = \sum_{a \in \mathcal{A}} z^a$.

Notice that the square of this series is:

$$f(z)^2 = \left(\sum_{a \in \mathcal{A}} z^a \right) \left(\sum_{b \in \mathcal{A}} z^b \right) = \sum_{n=0}^{\infty} r'_{\mathcal{A}}(n) z^n$$

where $r'_{\mathcal{A}}(n)$ is the number of solutions to $a + b = n$ with $a, b \in \mathcal{A}$ without the restriction $a \leq b$.

We have $r'_{\mathcal{A}}(n) = 2r_{\mathcal{A}}(n)$ if n is not twice an element of $\mathcal{A}$, and $r'_{\mathcal{A}}(n) = 2r_{\mathcal{A}}(n) - 1$ otherwise. In all cases, $r'_{\mathcal{A}}(n) \leq 2K$.

Thus, the function $f(z)^2$ is bounded on the real interval $z \in (0, 1)$ by:

$$f(z)^2 \leq \sum_{n=0}^{\infty} 2K z^n = \frac{2K}{1 - z}$$

This upper bound provides an explicit limit on the asymptotic behavior of $f(z)$ as $z \to 1^-$. $\square$

4 Architecture for Autoformalization

The code below illustrates the corresponding Lean 4 type structure.

```
import Mathlib.Data.Nat.Basic
import Mathlib.Data.Set.Finite

def r_A (A : Set Nat) (n : Nat) : Nat :=
  (Set.filter (fun p : Nat \times Nat => p.1 \in A \and p.2 \in A \and
    p.1 \le p.2 \and p.1 + p.2 = n) Set.univ).toFinset.card

def IsAdditiveBasisOrder2 (A : Set Nat) : Prop :=
  \exists N0 : Nat, \forall n \ge N0, r_A A n \ge 1

theorem erdos_turan_conjecture (A : Set Nat) (h : IsAdditiveBasisOrder2 A) :
  \forall K : Nat, \exists n : Nat, r_A A n > K := by
  sorry
```